

Tianyou Lai

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Research Profile

Reliable learning and evaluation under data heterogeneity, model unlearning, and context-dependent behavior; human–AI interaction and the legitimacy of LLM judgments. Applications span clinical imaging, industrial inspection, time-series forecasting, and engineering validation in energy systems.

Education

Carnegie Mellon University

Aug. 2026 – Dec. 2027 (expected)

M.S. in Artificial Intelligence Systems Management

Lanzhou University

Sep. 2021 – Jul. 2025

B.S. in Data Science & Big Data Technology | GPA: 87.34/100

The University of Texas at Austin

Jan. 2024 – Feb. 2024

Software Engineering Exchange Program | GPA: 4.0/4.0

Published and Accepted Papers

- [1] Yue, W.^{*}, **Lai, T.^{*}**, Luo, J., Mao, Q., Wang, Z., Ning, Z., and Li, Q. “TRACE: Training-time Report-guided and Clinically Ordered Concept Editing.” ACM MM 2026, accepted.
- [2] Zhou, H.^{*}, **Lai, T.^{*}**, Hao, C., Yue, W., Yue, S., and Zhang, H. “FREDNet: A Frequency and Decomposed-Spatial Network for Industrial Defect Signal Detection.” ICASSP 2026.
- [3] Yue, W., **Lai, T.**, Mao, Q., Li, Q., and Camacho, D. “A Review of Federated Learning under Data Heterogeneity.” Expert Systems, 2026, 70271.
- [4] Zhou, Y., **Lai, T.**, and He, X. “Deep Learning-Based Inverse Design of Broadband Metasurface Polarization Converter.” AIP Advances, 2025, 15, 075235.
- [5] Yue, S., Zhao, Z., **Lai, T.**, and Zhang, J. “A Modularity-Enhanced Echo State Network for Nonlinear Wind Energy Predicting.” Energies, 2025, 18(7), 1858.

^{*} Equal contribution.

Preprints

- [1] **Lai, T.^{*}**, Yue, W.^{*}, Zhou, J., Hao, C., Chang, L., Mao, Q., Niu, Z., and Li, Q. “ReLaMix: Residual Latency-Aware Mixing for Delay-Robust Financial Time-Series Forecasting.” arXiv, 2026.

^{*} Equal contribution.

Research Experience

Learning Systems and Human–AI Interaction

Federated and Heterogeneous Learning

Sep. 2024 – Present

Led the framework of a survey on data heterogeneity (Expert Systems, 2026); developed frequency-adaptive semantic distillation. Recent co-authored submissions cover structured partitioning for heterogeneous edge devices (FedSAP, AAAI 2027), semantic–appearance correction for personalized federated vision–language models (AAAI 2027), and verifiable questioning for federated language models (FedSocratic, ICLR 2027).

Unlearning, Context Effects, and Model Evaluation

2026

First author, *Reasoning after Unlearning* | Submitted to ICLR 2027

- Studied whether improved factual answering after unlearning reflects contextual assistance, answer exposure, or use of supplied answer information; evaluated original, unlearned, and target-excluded retrained Llama checkpoints on TOFU.
- Compared direct answering, matched-length controls, natural reasoning, and answer-masked context; used fixed-note masking and identity substitution to examine answer-information use.

Related co-authored ICLR 2027 submissions examine auxiliary supervision in multimodal unlearning and misleading comparison feedback in model selection.

Human-AI Interaction and LLM Judges

2026

When Outputs Become Outcomes (first author; submitted to CHI 2027): studied content agreement, acceptance of a judge's authority, and subsequent action in livestream games and academic reviewing; examined source attribution and human mediation through observational, interview, and presentation-comparison methods.

Collaborative Multi-Agent Reinforcement Learning Research

2026

Co-authored an AAAI 2027 submission on gradient routing and optimizer-dependent maintenance of learned cooperation.

Applied Machine Learning

Healthcare and Clinical Language

Nov. 2024 – Present

- **TRACE** (co-first author; Southern Medical University): BI-RADS-aligned breast ultrasound concepts with report supervision during training and image-only inference; ACM MM 2026, accepted.
- **Spike/EEG detection** (algorithm lead; Peking University): RT-DETR and spatiotemporal graph methods for juvenile/adult rat electrophysiology data.
- **TCM text augmentation** (independent honors thesis, 2025): dataset curation, prompting, and fine-tuning; text length reduced up to 44.1%, terminology tokens increased 4.4-fold, diagnostic accuracy up to 81.3%.

Industrial Inspection and Physical Inverse Design

Sep. 2024 – Jul. 2025

- **FREDNet** (co-first author): frequency/spatial modeling, data curation, training, ablations, and cross-domain evaluation on NEU-DET/GC-10; reported mAP@0.5 improvement of 4.8%; ICASSP 2026.
- **Metasurface inverse design** (second author): CNN-Transformer regression with spectral data and Gaussian-noise augmentation; mean MSE 0.00331; AIP Advances (2025).

Time-Series Forecasting and Energy Systems

Dec. 2024 – Present

- **ReLaMix** (co-first author): residual bottleneck mixing for delayed financial observations; PAXGUSD/BTCUSD evaluation with non-overlapping temporal segments; arXiv preprint (2026).
- **Artificial Intelligence for Critical Heat Flux in Energy Systems: From Predictive Accuracy to Engineering Use** (review in preparation; planned submission to *Energy*): synthesizes prediction and sensing studies, audits a public CHF benchmark, and links validation, uncertainty, and deployment boundaries to engineering decisions.
- **Wind energy prediction** (third author): modular echo state networks and turbine-data training; best R^2 0.9905; *Energies* (2025).

Industry Experience

Sichuan DeepGraph Intelligent Technology

Sep. 2024 – Feb. 2025

Data Science Intern | Prepared and validated Python datasets; queried graph relationships with Ultpa and supported OpenStack deployment.

Lanzhou Bronze Ding Intelligent Technology

Mar. 2024 – Sep. 2024

Data Engineer Intern | Processed 20+ datasets and 10,000+ labeled records using Python/SQL; built seven reports and deployed MySQL/Hadoop/Spark on a private cloud.

Guotai Junan Futures

Mar. 2024 – Apr. 2024

Quantitative Research Assistant | Extended a Python futures trading system; debugged CTP-API integrations and implemented TWAP/VWAP execution.

Zhongzheng Tiantong CPA

Jul. 2023 – Sep. 2023

Tax and Business Consulting Intern | Analyzed financial and risk data in Python/SQL and Tableau.

Feitian Technologies

Jul. 2022 – Jan. 2023

Product Data Engineer Intern | Maintained product data and supported C/C++ feature testing and performance reports.

Skills, Honors, and Service

Programming and AI: Python, SQL, C/C++, Java; PyTorch, TensorFlow; LLM fine-tuning and evaluation, federated learning, computer vision, time-series modeling.

Data and systems: MySQL, Hadoop, Spark, Hive, HDFS, Ceph, OpenStack, Docker, Kubernetes, Linux.

Research and presentation: Benchmarks, ablations, cross-domain evaluation; Matplotlib, Seaborn, Tableau, PowerPoint. Lanzhou University academic, internship, and exchange scholarships; undergraduate honors thesis (2025).

Reviewer: ICASSP 2027, ACM MM 2027, IJCNN 2026. English: TOEFL 103; GRE 339.